Parking Lot Management System

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1. Overview

The Parking Lot Management System is a web-based application developed using PHP, MySQL, HTML, CSS, and JavaScript. The purpose of this system is to manage parking operations efficiently by tracking vehicles, parking spots, tickets, and payments.

This system allows users to assign vehicles to parking spots, generate tickets, process payments, and manage parking availability in real time.

Aglionby designed the database structure using MySQL. This involved creating tables, setting primary keys, and establishing relationships between tables such as vehicles, parking records, and payments. The design ensures that each vehicle and parking activity is properly tracked and linked.
John developed the user interface using Visual Studios. The system includes forms that allow users to enter vehicle details upon arrival, View parking information, Process payments during exit. John implemented the backend logic using PHP. This includes handling form submissions, communicating with the MySQL database, retrieving and storing data, calculating parking fees based on time parked

2. Objectives

- To manage parking spots and vehicle assignments efficiently
- To track entry and exit times of vehicles
- To generate and manage parking tickets
- To process payments and maintain transaction records
- To demonstrate the use of SQL joins for data retrieval

3. Technologies Used

- PHP (server-side scripting)
- MySQL (database management)
- HTML, CSS (frontend design)
- JavaScript (optional interactivity)
- XAMPP/WAMP (local server environment)

4. System Features

4.1 Dashboard

The dashboard displays:

- List of vehicles
- Parking spots with availability status
- Tickets and payment information Parking spots are visually represented:

- Available spots are shown in green
- Occupied spots are shown in red Ticket statuses:

- PAID tickets are shown in green
- UNPAID tickets are shown in orange

4.2 Create Ticket

This feature allows assigning a vehicle to an available parking spot.

Steps:

1. Select a vehicle from the dropdown menu
2. Select an available parking spot
3. Click “Create Ticket”

Result:

- A new entry is created in the EntryLog table
- A ticket is generated in the Ticket table
- The selected parking spot is marked as occupied

4.3 Complete Ticket / Payment

This feature is used when a vehicle leaves the parking lot.

Steps:

1. Select the ticket ID
2. Enter the payment amount
3. Choose a payment method
4. Click “Complete Ticket”

Result:

- Ticket status is updated to PAID
- Exit time is recorded in EntryLog
- Parking spot becomes available again
- Payment record is stored in the Payment table

4.4 Admin Management

Manage Vehicles

- Add new vehicles (plate number and type)
- Delete existing vehicles

Manage Parking Spots

- Add new parking spots (number, type, availability)
- Delete parking spots

5. Database Design

The system uses the following tables:

- Vehicle
- ParkingSpot
- EntryLog
- Ticket
- Payment

Relationships

- A vehicle can have multiple entry logs
- Each entry log is linked to a parking spot
- Each ticket is associated with an entry log
- Each payment is linked to a ticket

6. SQL Joins

Query 1: Vehicle Parking Details

This query retrieves vehicle information along with parking spot and entry/exit times.

```
SELECT v.plate_number, ps.spot_number, el.entry_time, el.exit_time
FROM EntryLog el
JOIN Vehicle v ON el.vehicle_id = v.vehicle_id
LEFT JOIN ParkingSpot ps ON el.spot_id = ps.spot_id;
```

Query 2: Ticket and Payment Details

This query retrieves ticket information along with vehicle details and payment records.

```
SELECT t.ticket_id, v.plate_number, t.status, p.amount, p.payment_time
FROM Ticket t
JOIN EntryLog el ON t.log_id = el.log_id
JOIN Vehicle v ON el.vehicle_id = v.vehicle_id
LEFT JOIN Payment p ON t.ticket_id = p.ticket_id;
```

7. Security Features

- Use of prepared statements to prevent SQL injection
- Input validation for form data
- Proper handling of database queries

8. How to Run the Application

1. Install XAMPP or WAMP
2. Start Apache and MySQL
3. Import the database using phpMyAdmin 4. Place project folder in the htdocs directory
5. Open browser and go to: http://localhost/parking_lot_project/index.php

Parking Lot Management System

Vehicles

ID	Plate Number	Type
1	QWE1234	Car
2	RTY5678	Truck
3	UIO9009	Motorcycle

Parking Spots

ID	Spot Number	Type	Available
1	P1	Car	No
2	P2	Car	Yes
3	P3	Truck	No
4	P4	Motorcycle	Yes

Tickets & Payments

Ticket ID	Plate Number	Status	Payment Amount	Payment Time
1	QWE1234	PAID	20.00	2026-04-02 10:05:00
2	RTY5678	UNPAID	-	-
3	UIO9009	UNPAID	-	-

Create Ticket

Vehicle: UIO9009 (Motorcycle) Spot: P2 (Car) Create Ticket

Complete Ticket / Payment

Ticket: 2 - RTY5678 Amount: Payment Method: Cash Complete Ticket

Manage Vehicles

Add Vehicle

Plate Number: Vehicle Type: Car Add Vehicle

Existing Vehicles

ID	Plate Number	Type	Action
1	QWE1234	Car	Delete
2	RTY5678	Truck	Delete
3	UIO9009	Motorcycle	Delete

Manage Parking Spots

Add Spot

Spot Number: p1 Spot Type: Truck Available: Add Spot

Existing Spots

ID	Spot Number	Type	Available	Action
1	P1	Car	No	Delete
2	P2	Car	No	Delete
3	P3	Truck	No	Delete
4	P4	Motorcycle	Yes	Delete

10. Conclusion

The Parking Lot Management System successfully demonstrates how to build a full-stack web application using PHP and MySQL. It efficiently manages parking operations and showcases the use of SQL joins, and database relationships.